Question # 10 of 10 (Start time: 09:41:07 PM, 04 July 2024)

The volume of the solid bounded by planes x=a and x=b with cross-sectional area A(x) perpendicular to the x-axis is

Select the correct option

Re

○	$V = \int_a^b A(x)dy$
○	$V = \int_a^b A(y)dy$
○	$V = \int_a^a A(y)dx$
●	$V = \int_a^b A(x)dx$

Click to Save Answer & Mo

Question # 9 of 10 (Start time: 09:40:21 PM, 04 July 2024)

The volume of a cylinder is the area of a cross section of the cylinder multiplied by the _____ of the cylinder.

Select the correct option

○	Diameter
○	Radius
○	Base
●	Height

Click to Save Answ

Question # 8 of 10 (Start time: 09:39:31 PM, 04 July 2024)

For a function $f(x)$ that is decreasing on the interval $[a, b]$, how would this affect the volume of the solid of revolution?

Select the correct option

Rel

- ○ No effect on volume.
- ○ Volume increases.
- ◉ Volume decreases.
- ○ Cannot be determined.

Click to Save Answer & Mov

Question # 7 of 10 (Start time: 09:38:13 PM, 04 July 2024) **Total Marks:** 1

$$If\ f\ continuous\ on\ [a,b]\ and\ F(x)=\int_a^x f(t)dt,\ then$$

Select the correct option

Reload Math Equations

○	none of these
○	$F^{/}(t)=f(x)\ on\ \ [a,b]$
○	$F^{/}(x)=f(t)\ on\ \ [a,b]$
○	$F^{/}(x)=f(x)\ on\ \ [a,b]$

Click to Save Answer & Move to Next Question

9:38 PM
vulms.vu.edu.pk/

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 6 of 10 (Start time: 09:37:24 PM, 04 July 2024)

What is the purpose of using the washer method instead of the disk method?

Select the correct option

○	To simplify the integration process
○	To increase accuracy
●	To handle shapes with holes.
○	To reduce the number of slices

9:37 PM
vulms.vu.edu.pk/

Question # 5 of 10 (Start time: 09:36:31 PM, 04 July 2024)

The value of $\int_1^x y^2 dy =$ _____.

Select the correct option

- $\frac{y^3}{3} - \frac{1}{3}$
- $\frac{x^3}{3} - \frac{1}{3}$

9:36 PM
vulms.vu.edu.pk/

Question # 4 of 10 (Start time: 09:35:46 PM, 04 July 2024)

The expressions $x^2 + x, x^2 + x + 5, x^2 + x - 3$ have the same ..

Select the correct option

○	Anti-derivative
◉	Derivative

9:35 PM
vulms.vu.edu.pk/

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 3 of 10 (Start time: 09:35:06 PM, 04 July 2024)

When using the technique of slicing to find the volume of an irregular solid, what is the basic idea?

Select the correct option

◉	Divide the solid into slices and sum their volumes.
○	Divide the solid into cylinders and sum their volumes.
○	Divide the solid into cubes and sum their volumes.
○	Measure the distance between slices.

Click to Sa

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 2 of 10 (Start time: 09:34:34 PM, 04 July 2024)

If the upper limit of Definite Integral is equal to its lower limit,then the value of Definite Integral will be ___

Select the correct option

○	Same
○	1
●	Zero
○	None of the above

9:34 PM
vulms.vu.edu.pk/

MTH101 - Calculus And Analytical Geometry (Quiz No. 4)

Question # 1 of 10 (Start time: 09:33:48 PM, 04 July 2024)

If the radius of a sphere is doubled, how does the volume change?

Select the correct option

○	It becomes four times of the original volume.
◉	It becomes eight times of the original volume.
○	It becomes half of the original volume.
○	It remains the same of the original volume.

4:21 AM

vulms.vu.edu.pk/

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 10 of 10 (Start time: 04:19:44 AM, 01 July 2024)

What will be the value of $\int_0^1 e^x dx$?

Select the correct option

○	e^{c+1}
○	1
○	e
●	e^x

4:19 AM

vulms.vu.edu.pk/

Question # 9 of 10 (Start time: 04:18:35 AM, 01 July 2024)

Which of the following is the indefinite integral of f(x)=3/(x^2)

Select the correct option

○	(3/x)
○	None of these.
○	(1/4) x^ (4/3) +c
●	(-3/x) +c

Question # 8 of 10 (Start time: 04:17:51 AM, 01 July 2024) **Total Marks:** 1

Let $f(x)= x$ and $g(x)=2x$ are integrable functions over the interval $[-a, 0]$ for all $x\epsilon[-a, 0]$ then which of the following expressions is true for f and g?

Select the correct option

Reload Math Equations

- $\int_{-a}^{0} f(x)dx < \int_{-a}^{0} g(x)dx$
- $\int_{-a}^{0} f(x)dx \leqslant \int_{-a}^{0} g(x)dx$
- $\int_{-a}^{0} f(x)dx \geqslant \int_{-a}^{0} g(x)dx$
- $\int_{-a}^{0} f(x)dx > \int_{-a}^{0} g(x)dx$

Click to Save Answer & Move to Next Question

4:17 AM

vulms.vu.edu.pk/

Question # 7 of 10 (Start time: 04:17:04 AM, 01 July 2024)

Area of a rectangle whose width is 5 units and length is 6 units will be ....

Select the correct option

○	22 units
◉	30 square units
○	None of these
○	11 units

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 6 of 10 (Start time: 04:16:12 AM, 01 July 2024)

1+2+3+...+200 equals _____.

Select the correct option

○	21021
○	21220
◉	20100
○	20012

Question # 5 of 10 (Start time: 04:14:27 AM, 01 July 2024) **Total Marks:** 1

If $f(x)= x$ and $g(x)=2x$ are integrable functions over the interval $[0, a]$ for all $x \in [0, a]$, then which of the following expressions is true for f and g?

Select the correct option

Reload Math Equations

- $\int_0^a f(x)dx \leqslant \int_0^a g(x)dx$
- $\int_0^a f(x)dx > \int_0^a g(x)dx$
- (selected) $\int_0^a f(x)dx \geqslant \int_0^a g(x)dx$
- $\int_0^a f(x)dx < \int_0^a g(x)dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 4 of 10 (Start time: 04:13:51 AM, 01 July 2024)

Subdivide the interval [1, 5] into 'n' equally spaced subintervals then the width of each sub-interval is ----

Select the correct option

○	1/n
◉	4/n
○	5/n
○	5

Click to Sa

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 3 of 10 (Start time: 04:13:08 AM, 01 July 2024)

The integral of f(x) =sin(x) from x=0 to x=pi is .....

Select the correct option

◉	2
○	0
○	1
○	None of these

4:13 AM

vulms.vu.edu.pk/

MTH101 - Calculus And Analytical Geometry (Quiz No. 3) Qu

Question # 2 of 10 (Start time: 04:12:14 AM, 01 July 2024)

If we subdivide the interval [1,6] into n equal parts, then what will be the length Δx of each part?

Select the correct option

- (●) 5/n
- ( ) 1/n
- ( ) 6/n
- ( ) 7/n

Click to Save

MTH101 - Calculus And Analytical Geometry (Quiz No. 3) **Quiz Start Time: 04:10 AM, 01 July 2024**

Question # 1 of 10 (Start time: 04:10:04 AM, 01 July 2024) **Total Marks:** 1

For the function $f(x)=|x|-1$, if $f(-1)=f(1)=0$ then which of the following conclusion can be drawn about the point 'c' in the interval $[-1,1]$ such that $f'(c)=0$?

Select the correct option

○	c=0.5
○	No such 'c' exists
○	Every point in [-1,1] can be taken as 'c'
◉	c=0